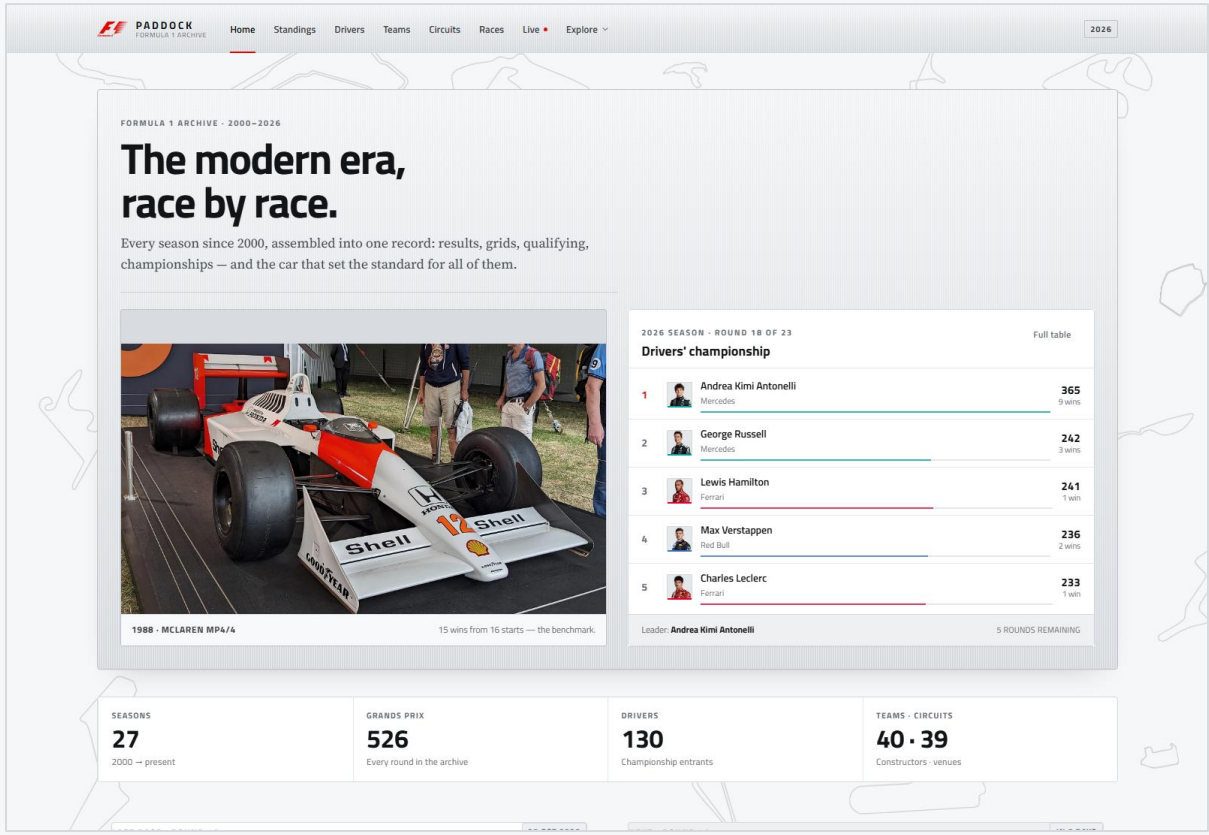
2000 → 2026

PADDOCK

Every season since 2000.
Every lap since 2023.

A brushed-silver Formula 1 archive built with Angular 21 and Material 3. The complete 2000–2026 dataset meets OpenF1 session replay, live timing and telemetry.



JUGAL PATEL · KRISH PATEL · PAAVAN SHARMA

24162581032 · 24162581017 · 24162581052

FET · SEMESTER 5

HOME · MP4/4 HERO · 2026 TOP FIVE · 526 RACES

THE BRIEF

The old app knew a lot. It showed very little.

A full data archive sat behind a front end that was hard to read, hard to navigate and out of date.

01

Fragmented experience

Nine pages with different visual rules, inconsistent spacing and no clear path from a season to a race to a driver.

02

Data without context

Raw result tables. No career totals, no championship picture, no comparison tools, no telemetry.

03

Stuck in the past

The archive stopped at 2024 and the current generation of cars had no live or replay layer.

THE GOAL

One coherent product: a calm, fast, 2000-onward archive with a real live layer — shipped end to end in one pass.

PRODUCT

One paddock, eleven screens.

Every route shares one silver system, and every table drills into a profile or a race.

27	526	10,954	11
SEASONS BUNDLED	ARCHIVED RACES	RACE RESULTS	SCREENS

/	**Home** MP4/4 hero, 2026 board, latest and next race	/races/:id	**Race detail** Classification, qualifying and season context
/standings	**Standings** Drivers' and constructors' tables, leader logic	/live	**Live timing** OpenF1 replay, timing tower and telemetry
/drivers	**Drivers** Searchable grid with career profile dialogs	/history	**History** Wikipedia-style eras, rivalries and records
/constructors	**Teams** Real logos or typographic crests, title years	/rivalry	**Rivalry** Any two drivers, career totals and shared seasons
/circuits	**Circuits** 39 layouts, searchable map and detail panel	/credits	**Credits** Every image, its source and its licence
/races	**Races** Season calendar, podiums and year switcher		

Brushed silver, one red signal.

A flat editorial system, not a dark dashboard. One theme, hairline rules, and type doing the work.

One theme, no toggle

Silver is the only surface. Dark mode, glass and glow were deleted, not restyled.

Flat panels

One .panel primitive: matte surface, hairline border, 2–4 px radius, one soft shadow.

Two typefaces

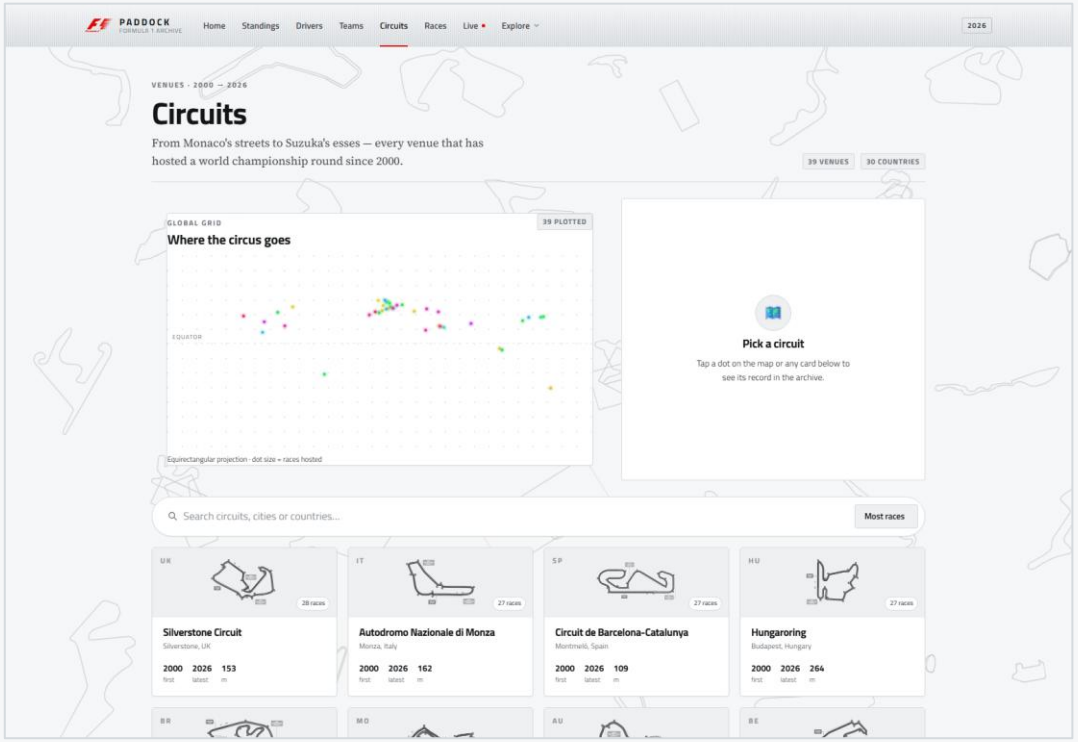
Titillium Web for UI and headlines, Source Serif 4 for the long-form history.

Red as a signal

F1 mark, active nav rule, P1 and one primary action per view. Never a background.

Maps as field

Thirty distinct black outlines at 20% ink cover the full canvas; ten on mobile, none under reduced motion.



ARCHITECTURE

A small front end with a clear spine.

Standalone Angular components, signals for state, and two services that own all data access.

SHELL	app.ts · navbar · router outlet · floating circuit layer
PAGES	home · standings · drivers · teams · circuits · races · race detail · live · history · rivalry · credits
SHARED UI	panel · page header · stat tile · driver avatar · team badge · track background · charts
CORE	CSVDataService · OpenF1Service · models · config · utils

KEY DECISIONS

— Lazy routes

Every page except Home is a separate chunk.

— Indexed CSVs

Parse once, derive careers and standings in memory.

— shareReplay

One HTTP request per archive file and OpenF1 call.

— Rate limited

3 req/s token bucket protects the OpenF1 free tier.

— Typed end to end

Strict TypeScript models for archive and OpenF1.

DATA

Two sources. One continuous timeline.

The archive gives depth and precision; OpenF1 gives recency and live detail.

2000 → 2026

Bundled archive

Ergast-style CSVs, pruned in the browser build.

- 130 drivers · 40 teams · 39 circuits
- 526 races · 10,954 results · 9,968 qualifying
- Final or current standings for all 27 seasons
- ~1.3 MB total, loaded locally, indexed once

→

2023 → TODAY

OpenF1 API

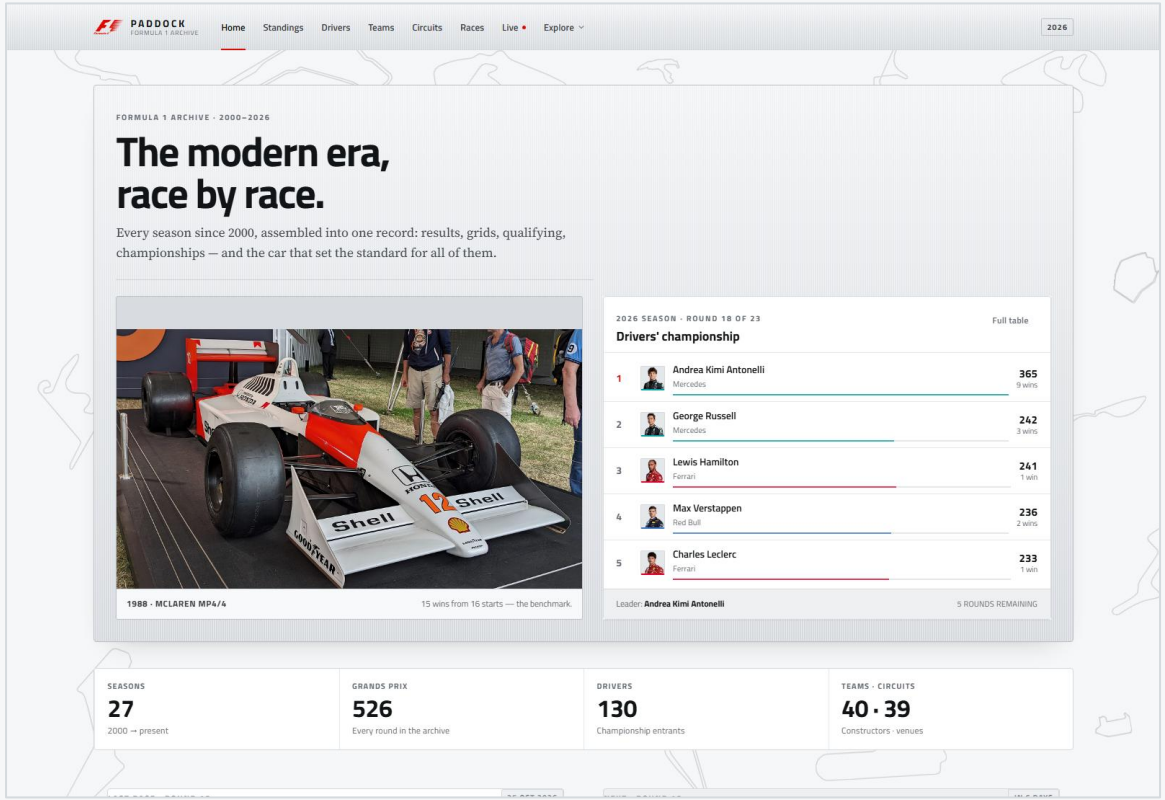
Session replay, timing and telemetry.

- Sessions · meetings · drivers · positions
- Intervals · laps · stints · pit stops
- Weather · race control · car data
- 3 req/s cap with cache + token bucket

2026 is reconciled to round 18 of 23 (26 October 2026). An unfinished season shows a leader — never a champion.

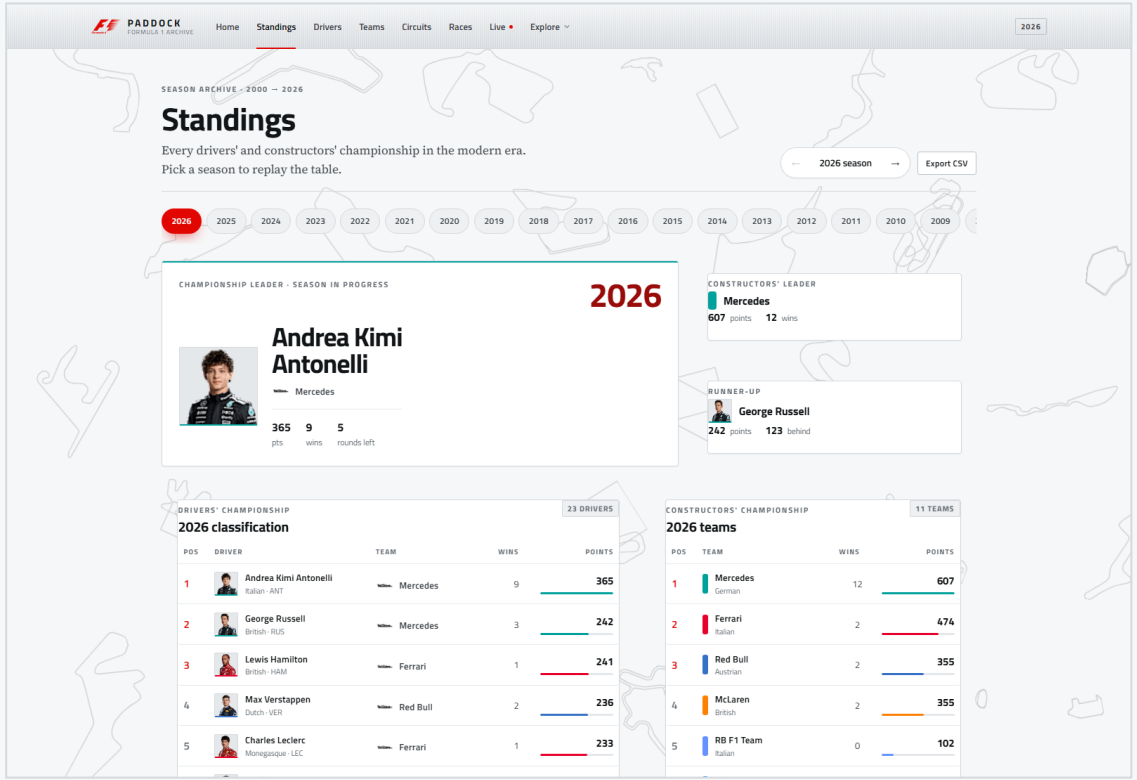
The season at a glance.

Home and standings share one hierarchy: context first, numbers second, detail one click away.



HOME

MP4/4 hero, 2026 top five, archive totals, latest and next race.

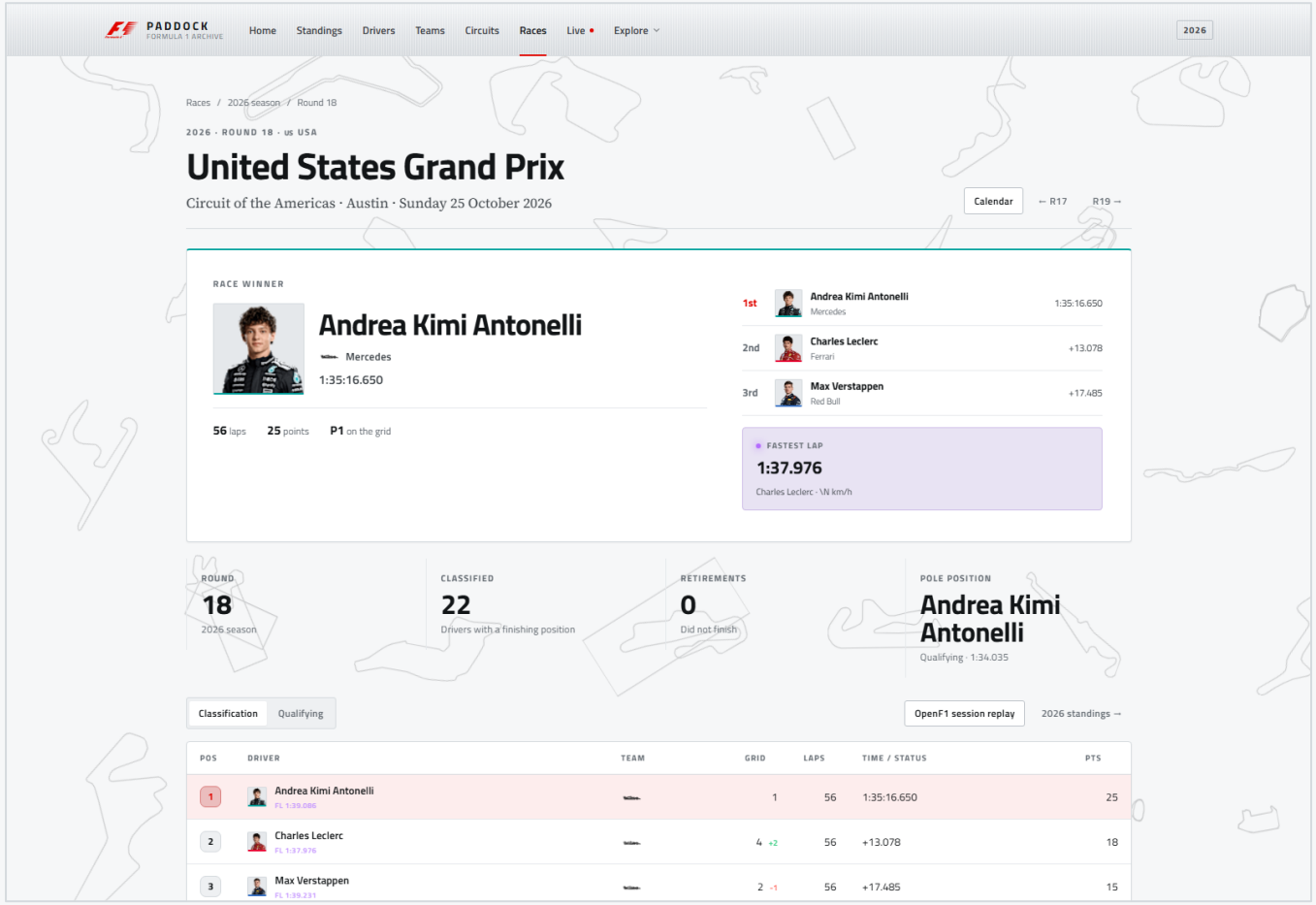


STANDINGS

Drivers' and constructors' tables, with a leader card for an unfinished season.

Every race gets a proper story.

A classification page turns a row in a CSV into a readable race report.



Winner hero

A 2 px team-colour rule, podium, race time and grid drop.

Fastest lap

Driver, lap time and top speed in one card.

Classification

Grid delta, laps, time/status, points and FL tag.

Qualifying

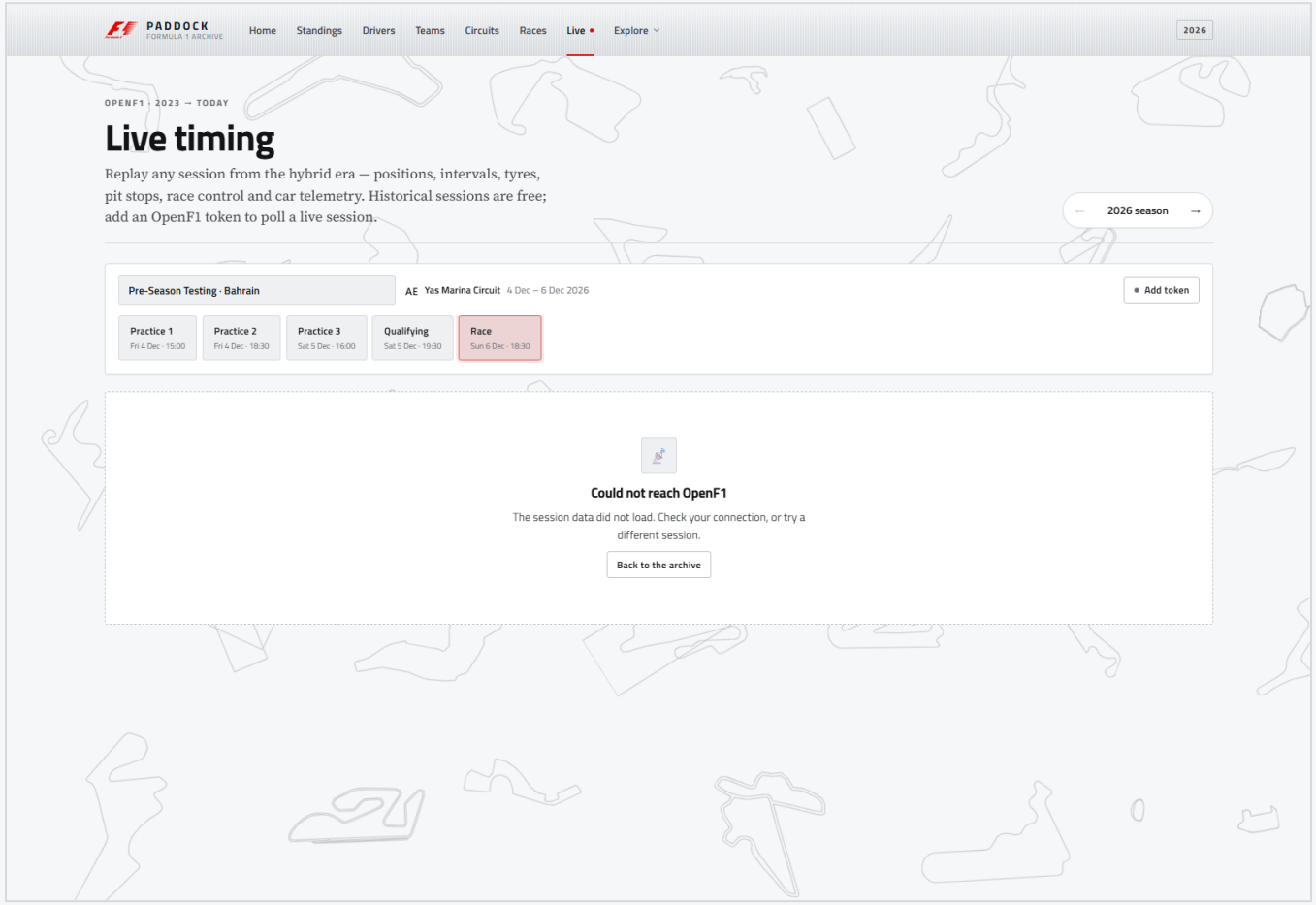
Q1 / Q2 / Q3 tab with the full session order.

Context

Circuit, season stats and previous/next round.

Replay today. Poll live when you have a token.

OpenF1 data drives a timing tower, race control, weather and a windowed telemetry trace.



Timing tower

Position, gap, interval, laps, last/best, tyre age, pit stops.

Race control

Flag-coloured message feed with lap and timestamp.

Telemetry

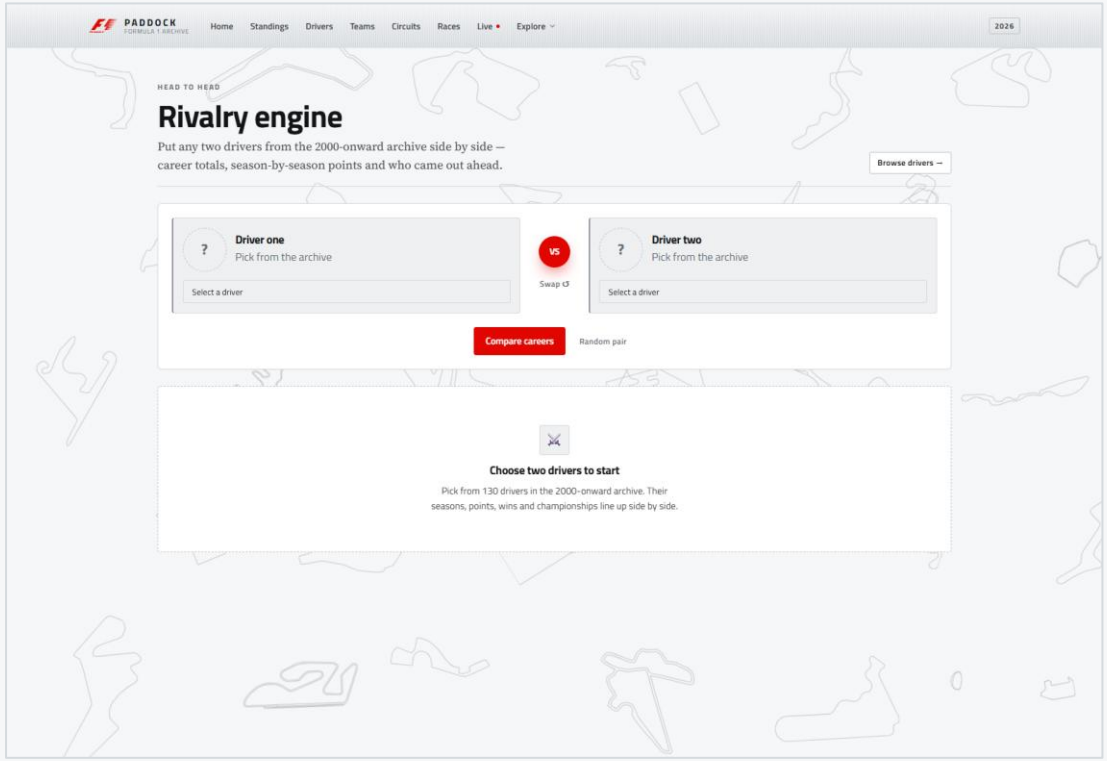
Speed, gear, DRS and gear changes for one lap only.

Safe by default

Polling off until a bearer token is stored locally.

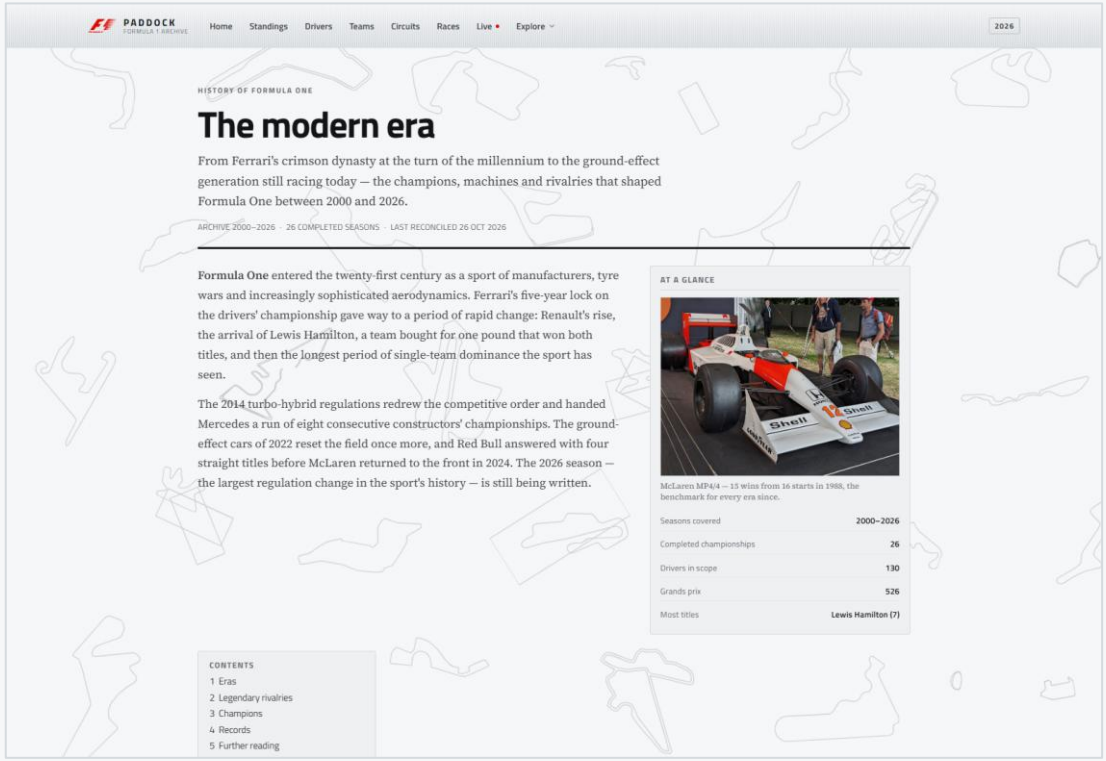
The archive becomes an argument.

Records, eras and head-to-head comparisons are computed from the same rows that power the tables.



Rivalry engine

Any two of 130 drivers. Career totals, points trajectory, shared seasons and a head-to-head score.

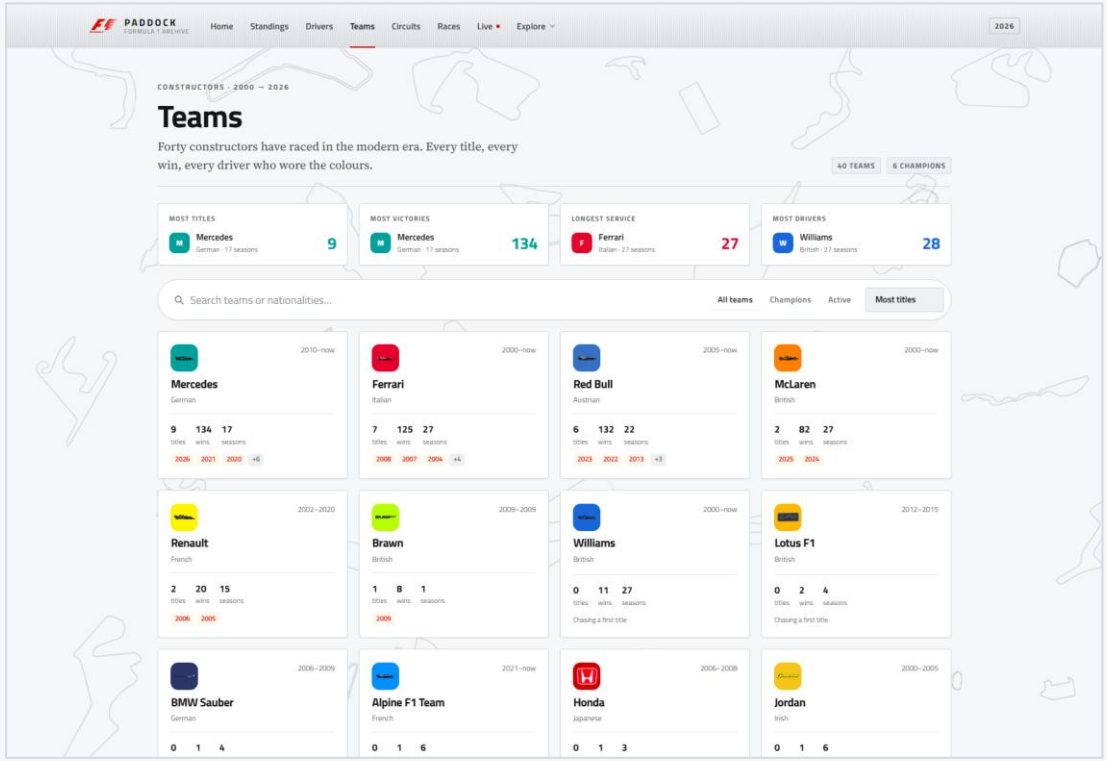


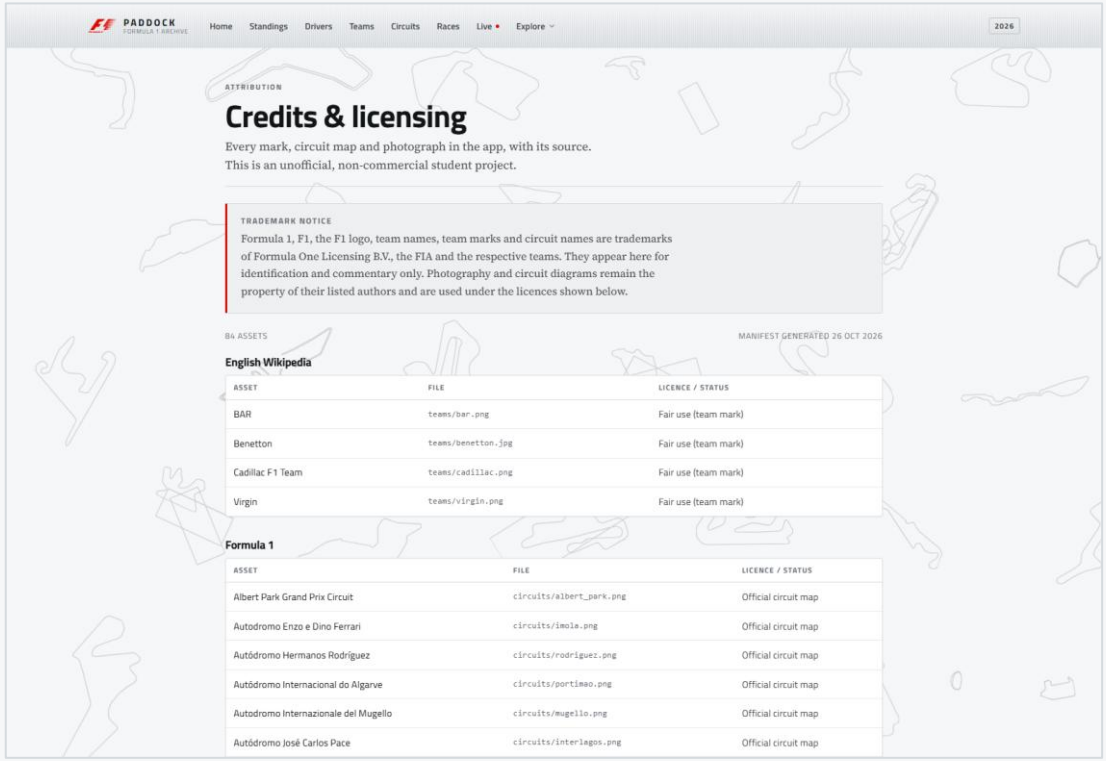
History article

Wikipedia-style lead, infobox, five eras, four rivalries, champions table and records.

Every team and every circuit has an image.

Real marks where they can be licensed, designed fallbacks where they cannot — and credit for all of it.





35 of 40

TEAM LOGOS

Five defunct teams render a typographic crest.

39 of 39

CIRCUIT MAPS

Every venue resolves to its official layout image.

28

DRIVER PORTRAITS

OpenF1 headshots; earlier drivers keep monograms.

9

ARCHIVE PHOTOS

MP4/4, Hunt–Lauda, Senna and more, credited in-app.

QUALITY

Fast, accessible, and honest about limits.

The build is lean, the interface is keyboard-friendly, and the app says when data is unavailable.

580 kB	154 kB	10	3 req/s
INITIAL RAW BUNDLE	INITIAL TRANSFER	LAZY ROUTES	OPENF1 BUDGET

PERFORMANCE

- Route-level code splitting
- Archived CSVs indexed once
- shareReplay on every request
- Budgets enforced in angular.json

ACCESSIBILITY

- Semantic headings and captions
- Labelled controls and tab states
- Visible focus and keyboard nav
- Reduced motion / transparency

RESILIENCE

- Skeletons instead of blank screens
- Designed empty and error states
- Crest fallback for missing logos
- Flag fallback for missing maps

DELIVERY

Clone. Install. Run.

No database, no API key and no build step for the bundled archive.

QUICK START

```
cd frontend
npm install
npm start

# production
npm run build

# full data refresh
npm run data:build
npm run assets:fetch
npm run outlines:build
```

WHAT SHIPS

App	11 routes · silver editorial · responsive
Archive	2000–2026 · 1.3 MB · pruned CSVs
Live	OpenF1 replay + optional token polling
Images	35 logos · 39 maps · 28 portraits · 9 photos
Docs	README.md · README.docx · README.pdf
Deck	f1-presentation.pptx · .pdf

NEXT

Known edges. Clear next moves.

The product is complete for this scope, and the boundaries are deliberate.

BOUNDARIES

2026 is still running

18 of 23 rounds are complete. The app labels the current leader and lists no 2026 champion.

Portraits start in 2023

OpenF1 only lists modern headshots. Earlier drivers keep the designed monogram.

Live needs a token

OpenF1's free tier is historical. Real-time polling requires a paid subscription.

Some marks cannot be licensed

Five defunct teams use a typographic crest instead of a logo.

NEXT

Sprint + lap data

Add sprint results and lap-by-lap archive data for deeper race analysis.

Favourites + sharing

Persist favourite drivers and shareable comparison links.

Caching layer

A small server cache for OpenF1 would make live pages instant for everyone.

More era photography

Extend the credited archive photo set beyond the nine key moments.

THANK YOU

The paddock is open.

Questions, laps, or a debate about the 2021 finale — all welcome.



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localhost:4200

openf1.org · angular.dev